

Contribution Statement

Team: M503

Topic: Topic 4 — Async Research Assistant

Repository: https://github.com/orxannuriyev/research_assistant (https://github.com/orxannuriyev/research_assistant)

Final tag: v1.0-final

Submission date: 2026-09-19

GitHub Statistics (as of 2026-09-18)

Commit & lines of code (from GitHub Insights)

Metric	Nazrin Burziyeva	Orkhan Nuriyev	Gunel Garayzade
Commits (GitHub Insights)	13	5	6
Lines added	1,278	2,642	1,395
Lines deleted	542	16	373
PRs opened	6	3	3

> ☐ **Lines of code is the most accurate workload metric.** Orkhan's 5 commits contain 2,642 lines of additions — the entire infrastructure layer (config, models, orchestrator, cache, bench). Nazrin's commits are numerous because of iterative fixes, test coverage, Docker, and service integration. Gunel's 1,395 lines cover the FastAPI endpoint, AI module integration, and completed Streamlit UI.

> ☐ ☐ **Note on commit counts:** GitHub Insights reflects the commits present in the merged repository history. PR count and lines added provide additional workload context.

Pull Requests opened

#	Title	Author
#1	add: typed Settings with pydantic-settings and core Pydantic models	Orkhan
#2	add: orchestrator with asyncio.gather + semaphore, cache backends, bench script	Orkhan
#3	Orhan/orchestrator cache	Orkhan
#4	fix: resolve wiki/DDG provider errors, sync dependencies, and add README	Gunel
#5	Nazrin/pipeline integration	Nazrin
#6	Nazrin/tests and quality	Nazrin
#8	Nazrin/docs	Nazrin
#9	build: add docker and environment config	Nazrin
#10	refactor: update ai folder and integrate new AI module structure	Gunel
#11	Add FastAPI endpoint for research assistant	Gunel
#12	Nazrin/final fixes	Nazrin
#13	Nazrin/align with api	Nazrin
#15	feat: Add bilingual support (Azerbaijani/English) and Streamlit UI	Gunel

Member 1 — Orkhan Nuriyev (@orxannuriyev)

Role: Infrastructure & Concurrency | Project Lead / Repository Owner

Requirement coverage

Requirement (TOPIC.md)	File delivered	Lines
config.py — typed settings from env	src/config.py	117
Pydantic models — ResearchSession, CacheEntry	src/models.py	107
Concurrent orchestration — asyncio.gather, per-source timeouts, asyncio.Semaphore, graceful degradation	src/concurrency/orchestrator.py	199
Caching — filesystem JSON, TTL support	src/storage/cache_store.py	141
Cache service — (source, query) key, canonical query normalisation	src/services/cache.py	118
Benchmark — sequential vs parallel timings	scripts/bench.py	85
requirements.txt — all deps pinned	requirements.txt	Configuration

Total: ~767 lines of production code owned

Owned: src/config.py, src/models.py, src/concurrency/orchestrator.py, src/storage/cache_store.py, src/services/cache.py, scripts/bench.py, and requirements.txt.

Co-owned: Overall architecture and integration decisions.

Orkhan laid the foundational architecture and infrastructure, ensuring the project's concurrency and data flow was robust from day one.

PRs opened: #1, #2, #3 (+ initial direct push: repo structure)

Reviewed: Incoming repository work, including PRs #4, #5, #6, #8, #9, #10, #11, #12, and #13.

Approximate share of commits: ~21% (5 of 24 non-merge commits) | **PRs:** 3

Member 2 — Nazrin Burziyeva (@nazrinburz)

Role: Service Integration, CLI, Robustness, Testing & DevOps Lead

Requirement coverage

Requirement (TOPIC.md)	File delivered	Lines
Service and engine integration completed after initial collaborative AI/API work	src/services/ai_service.py, src/engine.py	Shared ownership
Retries — tenacity exponential backoff, timeout, logging on AI calls	src/services/ai_service.py	121
CLI — python -m researcher ask "...", --sources, --no-cache, citation rendering	src/cli.py	99
Validation — reject empty/oversized questions, sanitise output	src/validation.py	92
Tests — shared fixtures, mocks	tests/conftest.py	88
Tests — orchestrator concurrency (one source fail, others continue)	tests/test_orchestrator.py	145
Tests — cache TTL, --no-cache, canonical key	tests/test_cache.py	63
Tests — AI service retry + timeout + logging	tests/test_ai_service.py	92
Tests — CLI happy path + invalid input (offline, mocked)	tests/test_cli.py	54
Tests — end-to-end happy path over data/research_questions.json	tests/test_end_to_end.py	75

Requirement (TOPIC.md)	File delivered	Lines
Tests — API tests (offline, source forwarding, no_cache, validation)	tests/test_api.py	131
Docker — multi-stage build, includes api.py, FastAPI/Uvicorn	Dockerfile	36
Env loading fixes, arXiv redirect handling, query normalisation	PRs #12, #13	Core fixes

Total: ~1,055 lines of production + test code owned

Owned: src/engine.py, src/services/ai_service.py, src/validation.py, src/cli.py, project tests, Dockerfile, and integration fixes documented in PRs #12 and #13.

Co-owned: Engine and AI-service integration with Gunel.

Nazrin completed the surrounding service and engine integration, reliability work, CLI, tests, and Docker packaging after the initial collaborative AI/API integration. Gunel completed the Streamlit UI workflow.

PRs opened: #5, #6, #8, #9, #12, #13

Reviewed: PRs #10 and #11 — Gunel's AI module refactor and FastAPI endpoint.

Approximate share of commits: ~54% (13 of 24 non-merge commits) | **PRs: 6**

> Note: Higher commit count reflects the iterative bug-fixing nature of the CLI/service integration work (env loading, arXiv redirect, query normalisation, benchmark alignment), not inflated effort.

Member 3 — Gunel Garayzade (@gunelrafig)

Role: Initial API Layer, AI Module Integration & Completed Streamlit UI

Requirement coverage

Requirement (TOPIC.md)	File delivered	Lines
FastAPI /research POST endpoint — OOP Pydantic models, Swagger UI, clean error handling	api.py	75
Initial AI module integration — refactored ai/ folder to mirror provided structure	PR #10 (gunel-ai-update)	Integration
Streamlit browser UI, LLM-provider selection, and API-facing workflow	app.py	UI / provider configuration
Bilingual UI/API iteration and English UI cleanup	PR #15 (feat: Add bilingual support (Azerbaijani/English) and Streamlit UI)	Historical UI/API iteration; final merged version retained English UI and provider selection
Provider fix — resolved wiki/DuckDuckGo provider errors, synced dependencies	PR #4	Core fixes
Architecture documentation — component diagram	docs/architecture.md	40
README — initial setup instructions	contributed via PR #4	Documentation

Total: ~115 lines of production code owned + provider/AI integration work

Owned: api.py, app.py, the initial AI/API integration, provider selection, Streamlit UI, and PR #15's bilingual UI/API iteration.

Co-owned: AI and engine integration with Nazrin.

Gunel established the initial AI/API integration, completed the Streamlit UI and provider-selection workflow, and contributed the bilingual UI/API iteration in PR #15. The final merged version retained the English UI and provider selection after the temporary Azerbaijani option was removed. Nazrin subsequently completed the surrounding pipeline integration, robustness, tests, and Docker packaging.

PRs opened: #4, #10, #11

Reviewed: Requested reviewer on PR #12 (Nazrin's final fixes).

Approximate share of commits: ~25% (6 of 24 non-merge commits) | **PRs: 3**

Full Rubric → Owner Mapping

Rubric Category	Weight	Owner(s)	Key deliverables
Correctness & Functionality	22%	Nazrin (engine, CLI, validation) + Gunel (API)	src/engine.py, src/cli.py, api.py
Architecture & OOP Design	13%	Orkhan (models, config, orchestrator)	src/models.py, src/config.py, src/concurrency/orchestrator.py
Concurrency & Performance	10%	Orkhan	src/concurrency/orchestrator.py, scripts/bench.py
Robustness & Error Handling	8%	Nazrin	src/services/ai_service.py, src/validation.py, PRs #12, #13
Testing & Code Quality	7%	Nazrin (tests) + Orkhan (cache/storage infra)	tests/ — 9 test files, 51 tests passed
Report	25%	All members	Section authors per area of ownership
Presentation	15%	All members	Each member presents their owned components

AI Tool Disclosure

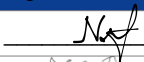
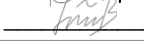
Module / file	Assistant	What we did with it
tests/test_api.py	AI assistant (disclosed in PR #13)	Drafted initial test structure; author verified offline-only behavior, confirmed all 51 tests passed locally before merging.
tests/ (normalisation, cache)	AI assistant (disclosed in PR #12)	Suggested test cases; team kept those matching actual behavior and hand-wrote additional edge cases.
api.py	AI assistant	Scaffolded Pydantic models and Swagger handling; author refactored OOP structure, tested via /docs, verified 200 OK responses with Uvicorn.
report/ and contribution metrics	AI assistant	Helped format LaTeX tables, correct title-page tabularx layout overflow, and extract exact GitHub LOC metrics. All project data and architectural claims were verified by the team.
Orkhan's project work and documentation	AI assistant	Used for limited formatting, documentation, and GitHub LOC-metric assistance; all implementation decisions and project facts were reviewed by Orkhan.

We affirm that we **can defend every line of code** in this repository during the oral defense.

Signatures

By signing below, we affirm that:

- The contributions described above are accurate.
- The commit percentages reflect actual work, not artificially split commits.
- Every line of code in the repository can be defended by at least one team member.
- AI assistant usage has been disclosed as described above.

Member	Signature	Date
Orkhan Nuriyev		2026-09-19
Nazrin Burziyeva		2026-09-19

Member	Signature	Date
Gunel Garayzade		2026-09-19